TEXORA · LECTURE SHEET

Synthesis and Characterization of Nanomaterials

The nanomaterials textiles actually use — carbon nanotubes, fullerenes, and the metal and metal-oxide nanoparticles (silver, silica, titania, zinc oxide, magnesia) — how they are made, how XRD, AFM, SEM/TEM and DLS prove what was made, and where nanotechnology serves the oldest textiles of all: museum collections.

TE-4206 · TECHNICAL TEXTILE AND NANO-TECHNOLOGY · TOPIC 7 · LEVEL 4 TERM 2

1. The materials, one by one

1.1 Carbon nanotubes (CNT)

Rolled graphene sheets forming tubes of nanometre diameter. Single-walled (SWCNT) and multi-walled (MWCNT) forms; outstanding axial strength and modulus, electrical conductivity from metallic to semiconducting depending on the roll geometry (chirality). **Synthesis:** arc discharge, laser ablation, and above all chemical vapour deposition (CVD) — hydrocarbon gas over a hot metal catalyst. **Textile relevance:** conductive and sensing yarns, composite reinforcement, heating elements; limits are cost, dispersion difficulty, and ensuring the tubes stay bound in the product.

1.2 Fullerenes

Closed carbon cages — C_{60} (“buckyball”) is the emblem, a truncated icosahedron of 60 carbon atoms. Made in carbon arcs and combustion processes. Properties: electron-accepting, radical-scavenging, photoactive. Textile use is still mostly research (functional additives, sensing), unlike CNTs — a useful lesson that fame and industrial use are different things.

1.3 Metal and metal-oxide nanoparticles

MATERIAL	COMMON SYNTHESIS	FUNCTION IN TEXTILES	NOTES
Nano silver (Ag)	Chemical reduction of $AgNO_3$ (citrate, borohydride, plant extracts); also physical routes	Antimicrobial, anti-odour, conductive inks	Works by Ag^+ release — rate rises as size falls (Topic 6)

Nano silica (SiO_2)	Sol–gel (Stöber process), flame hydrolysis	Surface roughness for water repellency, matting, carrier for other actives	Amorphous; chemically gentle
Nano titania (TiO_2)	Sol–gel / hydrothermal from titanium alkoxides or $TiCl_4$	UV absorption, photocatalytic self-cleaning	Anatase is the photocatalytic crystal form; rutile the UV-screening pigment form
Nano zinc oxide (ZnO)	Precipitation, hydrothermal, sol–gel	UV protection, antimicrobial	Less photocatalytically aggressive than TiO_2
Nano magnesia (MgO)	Precipitation/calcination, sol–gel	Antimicrobial, flame-retardant synergist, adsorbent	Alkaline surface chemistry drives its action

Table 1 — The syllabus's named oxides and silver. General synthesis logic is shared: control nucleation and growth (concentration, temperature, pH, stabilisers) and you control size; let particles collide unprotected and they agglomerate — the failure mode of the whole subject.

2. Worked example — how many particles are in a gram-scale synthesis?

WORKED EXAMPLE — REDUCING 100 ML OF 1 MMOL/L $AgNO_3$

Moles of Ag^+ = 0.100 L × 1 × 10^{-3} mol/L = 1.0 × 10^{-4} mol.

Mass of silver = 1.0 × 10^{-4} × 107.87 g/mol ≈ **10.8 mg**.

If all of it forms spheres of diameter 20 nm (radius 1.0 × 10^{-6} cm, ρ(Ag) = 10.49 g/cm³):

volume per particle = (4/3)π(1.0 × 10^{-6})³ = 4.19 × 10^{-18} cm³; mass per particle = 4.19 × 10^{-18} × 10.49 ≈ 4.39 × 10^{-17} g.

Number of particles = 0.0108 g / 4.39 × 10^{-17} g ≈ **2.5 × 10^{14}**.

Interpretation: even a bench-scale synthesis is a population of hundreds of trillions — which is why characterisation reports *distributions* (mean, spread), never a single “size”, and why a small agglomerated fraction can dominate behaviour.

3. Characterization techniques

TECHNIQUE	PRINCIPLE	WHAT IT TELLS	LIMITS

XRD	X-rays diffract from crystal planes; Bragg's law nλ = 2d sinθ	Crystal phase (anatase vs rutile), crystallite size from peak broadening (Scherrer), purity	Blind to amorphous material; gives crystallite, not particle, size
SEM	Scanned electron beam; secondary electrons image the surface	Surface morphology, fibre coatings, particle shape and distribution on fabric	Surface only; samples must survive vacuum (and often coating)
TEM	Electron beam transmitted through an ultra-thin specimen	True particle size and shape down to atomic lattices; core–shell structures	Tiny sampling volume; demanding preparation
AFM	A sharp tip scans the surface, feeling height atom by atom	True 3D topography and particle height at nanometre resolution, no vacuum needed	Slow, small areas; tip artefacts
DLS	Fluctuations of scattered laser light from particles diffusing in liquid	Hydrodynamic size distribution in the actual liquid product — fast and practical	Measures agglomerates as particles; assumes spheres; poor with mixtures

Table 2 — The syllabus's four characterisation families. They are complementary, not interchangeable: XRD says what crystal, TEM says what size, DLS says what the customer's liquid actually contains.

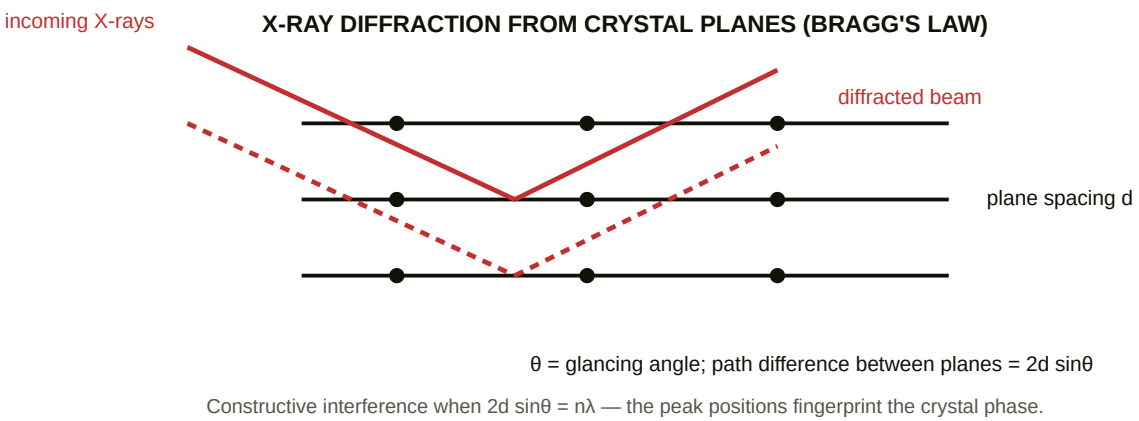


Figure 1 — Bragg diffraction: reflections from successive planes reinforce only at specific angles, set by the plane spacing d.

Worked example — reading an XRD pattern

WORKED EXAMPLE — ANATASE TiO_2, STRONGEST PEAK AT 2Θ = 25.6° (CU KA, Λ = 0.154 NM)

θ = 12.8°. Plane spacing: d = λ/(2 sinθ) = 0.154/(2 × sin 12.8°) = 0.154/0.443 ≈ **0.348 nm** — matching the anatase (101) plane (≈ 0.35 nm), so the product is the photocatalytic anatase form, not rutile.
Crystallite size from peak width (Scherrer, K = 0.9, FWHM β = 0.5° = 0.00873 rad):
D = Kλ/(β cosθ) = 0.9 × 0.154/(0.00873 × cos 12.8°) = 0.1386/0.00851 ≈ **16 nm**.
Interpretation: the mill can report “anatase, ~16 nm crystallites” — a claim with a method behind it, per Topic 6's discipline.

4. Applications in textiles and polymers — the map

Putting Sections 1–3 together: nano Ag → antimicrobial finishes and conductive printing; TiO_2 → UV protection and photocatalytic self-cleaning (Topic 9); ZnO → gentler UV/antimicrobial; SiO_2 → roughness-based water repellency and as a carrier; MgO → antimicrobial and flame-retardant assistance; CNTs → conductivity, sensing, reinforcement; nanoclays and oxides in **polymers** → barrier films and stiffer, more flame-resistant plastics (the nanocomposite mechanism of Topic 8). In every case the textile question is the same trio: *does it work, does it stay, is it safe* — function, durability, exposure.

5. Nanotechnology-based interventions in museum textiles

The syllabus's most unexpected phrase points at conservation science, where nanotechnology is used with unusual care — because the “product” is irreplaceable.

- **The problems.** Historic textiles suffer fibre degradation (acid hydrolysis of cellulose, weakened silk), soiling embedded over centuries, pest damage, and metal-thread corrosion. Conventional washing and solvents can destroy what they clean.
- **Nano-lime and alkaline nanoparticles.** Dispersions of calcium hydroxide nanoparticles in alcohol can penetrate degraded cellulosic fibres and deposits, neutralising acids and leaving a mild alkaline reserve — deacidification delivered gently, without immersing a fragile textile in water.
- **Nano-structured cleaning systems.** Microemulsions and gels loaded at nanoscale clean soiling selectively, with the cleaning fluid confined to the surface and removed completely — control being the whole point where the substrate is priceless.
- **Protective nano-coatings.** Thin silica-based and other nano-coatings are studied for protecting metal threads and fragile surfaces against tarnish and handling, designed to be minimal, documented and as reversible as conservation ethics demand.
- **The governing ethics.** Conservation interventions must be minimal, documented, and retreatable where possible; nanomaterials enter this field only when they are *gentler and more controllable* than

the conventional alternative — the opposite of marketing-led nano. It is the same discipline as Section 4's trio, sharpened: does it work, does it stay (only where wanted), and can it do harm over centuries?

Glossary

Chirality — the roll geometry of a carbon nanotube, deciding its electronic character. **Stöber process** — controlled sol–gel growth of uniform silica spheres. **Anatase / rutile** — the two common crystal forms of TiO_2. **Scherrer equation** — crystallite size from XRD peak broadening. **Hydrodynamic size** — the effective size DLS reports for a particle moving in liquid, coating and all. **Deacidification** — neutralising acids in degraded artefacts to slow further decay.

Key points

• CNTs (CVD-grown) and fullerenes are the carbon family; Ag, SiO_2, TiO_2, ZnO and MgO are the working oxide/metal set, each with a distinct function.
• Synthesis is nucleation-and-growth control; agglomeration is the standing enemy; a synthesis is a population — report distributions ($\approx 2.5 \times 10^{14}$ particles in one small bench example).
• XRD fingerprints phase and crystallite size (Bragg + Scherrer: anatase, ~16 nm in the worked example); TEM measures true size; SEM images surfaces; AFM feels topography; DLS sizes the product in its liquid.
• Applications follow the trio: function, durability, safety.
• Museum textiles show nanotechnology at its most disciplined: nano-lime deacidification, nano-structured cleaning, protective nano-coatings — minimal, documented, retreatable.

Practice

1. An XRD peak (Cu Kα, $\lambda = 0.154$ nm) appears at $2\theta = 27.4°$ with FWHM 0.4°. Calculate the plane spacing and the Scherrer crystallite size. ($\sin 13.7° \approx 0.237$, $\cos 13.7° \approx 0.972$; $0.4° = 0.00698$ rad.)
2. A supplier offers “nano TiO_2, 20 nm” for a self-cleaning finish. Which two techniques would you combine to verify the claim before purchase, and what would each prove?
3. Explain, in conservation terms, why a calcium hydroxide nanoparticle dispersion in alcohol can be safer for a fragile historic banner than an aqueous alkaline wash.

Answer key

1. $d = 0.154/(2 \times 0.237) \approx$ **0.325 nm** — consistent with rutile's main (110) spacing (≈ 0.325 nm), i.e. this is the rutile, not anatase, form. $D = 0.9 \times 0.154/(0.00698 \times 0.972) = 0.1386/0.00679 \approx$ **20 nm** crystallites.

2. **TEM** to measure the true primary particle size and shape (is it ~20 nm, and how much is agglomerated?), and **XRD** to prove the phase is anatase — because photocatalytic self-cleaning needs anatase; a 20 nm rutile would pass a size test yet fail the function. (DLS on the delivered dispersion is a good practical third check.)
3. The nanoparticles are small enough to penetrate the fibre structure and deposit where the acid damage is, and the alcohol carrier wets and evaporates without swelling the fragile fibres, without mobilising dyes and dirt through the textile, and without the mechanical stress of immersion and handling in water. The treatment is local, low-dose and controllable — the conservation criteria of minimal intervention and retreatability — whereas a bulk aqueous wash risks immediate physical loss of the very fibres it aims to save.
